

UTS-UAV: Context–Evidence Benchmarking for Compositional Operational Event Understanding in Low-Altitude UAV Inspection

Anonymous ICRA Submission

Abstract—Low-altitude UAV inspection requires recognizing operational events rather than merely naming visible objects: garbage may be floating or waterside, materials may be useful stock or construction waste, and smoke may indicate emission or fire. These decisions require scene context and small local evidence. We introduce UTS-UAV, a development snapshot toward a real-world benchmark with a compositional event ontology and two linked tasks: context–crop event recognition and context-level multi-evidence set prediction. A content audit reconciles 9,093 pair annotations across 45 present concepts into 7,279 exact-hash-unique wide contexts, 9,056 unique crops, and 9,082 unique context–crop pairs. Crucially, 1,389 contexts link multiple evidence crops and 269 contexts carry multiple event labels; we preserve this structure as a Context–Evidence Event Graph rather than misreporting every pair as an independent frame. We recover 1,549 unique class-level business record IDs, but zero can yet be deterministically joined to a local image. A normalized source-record/media/context gate therefore blocks the official split and acquisition-level claims instead of laundering filename dates into flights or sites. The benchmark separates a support-gated core from long-tail, open-vocabulary, and hard-negative tracks, with independent human-review ledgers and verified acquisition provenance as release gates. We further propose Context-Local Evidence Alignment and Relational Adaptation (CLEAR), a parameter-efficient VLM baseline with graph-neighbor rejection and a permutation-invariant set extension. Its protocol tests whether joint predictions truly require complementary views through single-view insufficiency, assigned-evidence deletion, prompt paraphrases, and crop-budget diagnostics. This draft reports verified audit facts and a frozen evaluation design; model results remain **TBD** until the fail-closed gate is ready and three-seed runs are complete.

I. INTRODUCTION

Routine UAV inspection is a robotics perception problem with an unusually large gap between what the camera sees and what an operator must decide. A wide aerial frame may contain roads, houses, waterways, vehicles, and work sites, but the actionable output is an *operational event*: a fire lane is obstructed, waste floats on water rather than lying on the bank, a rural building is under construction, or stored material is actually debris. These labels compose an entity, state, relation, and scene context. The evidence is often a small region, while the correct interpretation depends on the full scene.

Recent low-altitude vision-language resources have raised the scale bar. UAVBench/UAVIT-1M evaluates ten image/region tasks and supplies more than a million instructions [1]; SpatialSky-Bench covers 13 UAV spatial-intelligence subcategories with a million-sample training resource [2]; UAVReason unifies spatial/temporal reasoning with aerial generation in simulation [3]; and AirNav targets

real-world UAV vision-language navigation [4]. Consequently, “UAV imagery plus VLM fine-tuning” is no longer a sufficient contribution. What remains underrepresented is operationally grounded, fine-grained event understanding in repeated real inspection flights, where definitions are policy-facing, classes form explicit confusion sets, and the system must know when evidence is insufficient.

Our data expose this problem in a form that generic benchmarks do not. Each visible annotation links a wide *context view* to a local *evidence view*; repeated exact contexts may contain several crops and several valid events. The catalog also contains closely related concepts such as floating garbage, waterside garbage, unclean water, and algal bloom; or rural construction, accumulated building materials, and construction waste. These cases break object-name shortcuts and make naive pair-level splitting invalid: crops from the same scene, near-duplicate frames, and repeated flights can leak nearly identical evidence across train and test.

We reorganize the project around three ideas. First, a five-factor Operational Event Graph represents every label as a composition of domain, entity, state, relation, and context, with explicit neighbor edges for confusable concepts. Second, a Context–Evidence Event Graph preserves the empirical many-to-many relation among a wide context, its evidence crops, and its event set; this supports both pair recognition and context-level set prediction. Third, the benchmark is support-gated and leakage-aware, while CLEAR contrasts each positive event against graph neighbors and aggregates a variable-size evidence set without imposing crop order.

This paper makes three contributions:

- We introduce UTS-UAV, a real low-altitude inspection benchmark with 9,093 reconciled pair annotations over 7,279 exact-hash-unique contexts and a 67-concept operational catalog. We publish audit gates, content-group identities, and versioned adjudication ledgers instead of treating pair rows or raw records as independent images.
- We contribute an Operational Event Graph, a many-to-many Context–Evidence Event Graph, evidence cards, and leakage-resistant tasks spanning pair recognition, multi-evidence event-set prediction, hard negatives, open-vocabulary transfer, unseen locations, selective prediction, and explicit tests of whether a second view is necessary.
- We propose CLEAR, a Qwen3-VL-based parameter-efficient baseline that combines context and evidence with taxonomy-neighbor ranking and counterfactual

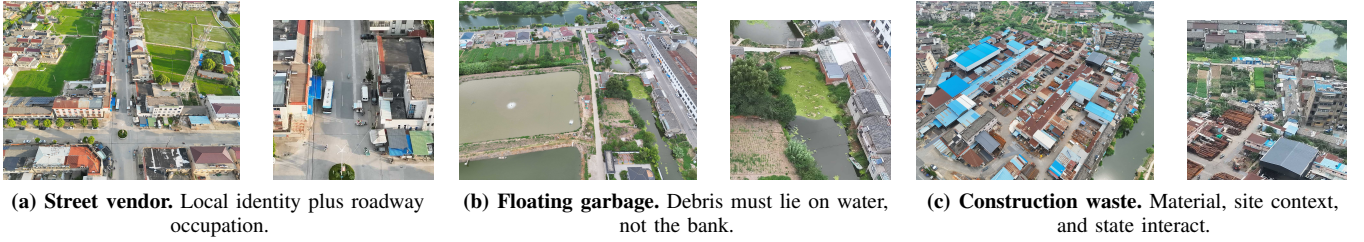


Fig. 1. Each annotation links a global context (left) to local evidence (right); exact-context grouping further joins multiple crops and events from the same scene. Operational labels require factor composition, not object naming. Images are internal examples pending release clearance.

rejection; its deterministic set extension exposes which crop supports each predicted event.

The manuscript is intentionally result-honest: verified snapshot statistics are included, whereas unresolved split counts and measurements remain **TBD**. The submission version must replace every placeholder with outputs from the frozen protocol in Section V.

II. RELATED WORK

a) UAV perception and event benchmarks.: VisDrone established detection and tracking under scale variation, occlusion, and camera motion [5]; UAV-Human expanded aerial understanding to actions, pose, re-identification, and attributes [6]. AIDER covers five broad emergency conditions [7], while ERA and CapERA provide 25 aerial video events and event captions [8], [9]. Newer resources are substantially broader: UAVBench/UAVIT-1M combines 10 benchmark tasks and 11 instruction-tuning tasks over real low-altitude imagery [1], SpatialSky-Bench evaluates 13 spatial-intelligence categories [2], UAVReason studies 22 spatial and temporal reasoning types in simulation [3], and AirNav addresses real-world aerial navigation instructions [4]. UTS-UAV does not compete on raw instruction volume; it targets policy-facing operational distinctions, paired wide/crop evidence, and group-disjoint evaluation from repeated inspection flights.

b) Open-world and language-defined events.: GeoChat and SkySense-O adapt vision-language models to remote-sensing interpretation and grounding [10], [11]; RegionGPT demonstrates the value of region-level captions and task-guided prompts [12]. LaGoVAD shows that anomaly meaning can change with a user-provided language definition and introduces multi-level semantic descriptions for open-world video anomaly detection [13]. This is closely related to operational inspection: the same object can be acceptable or actionable depending on its state, relation, and scene. We therefore evaluate class names, natural-language definitions, and definition-plus-exemplar prompts separately rather than assuming a fixed closed vocabulary.

c) Synthetic supervision and efficient adaptation.: LLaVA and ShareGPT4V established machine-generated instruction data and rich captions as scalable multimodal supervision [14], [15]. Unconstrained teacher text can omit small evidence or rationalize an incorrect label. We separate label-agnostic perception from label-conditioned rewriting, reject unsupported counterfactuals, and reserve human review

for high-risk strata. LoRA and QLoRA reduce trainable parameters [16], [17]. Our Qwen3-VL student [18] contributes context–evidence alignment and graph-structured supervision, not LoRA itself.

d) Dataset quality as part of the benchmark.: Repeated robotic observations make dataset curation inseparable from evaluation. UAVLight illustrates the value of repeatable geo-referenced flight paths and controlled time-of-day variation [19]; MMLandmarks shows how one-to-one instance correspondence across modalities enables auditable cross-view tasks [20]. A detector crop and its parent frame are not independent samples; neither are adjacent frames or repeat flights over the same site. Recent multimodal leakage analysis further shows why sample-level deduplication and higher-level source separation are distinct requirements [21]. UTS-UAV therefore treats source reconciliation, perceptual deduplication, provenance, group construction, class support, reviewer ledgers, and release versioning as benchmark components.

e) Multi-view necessity.: Multi-image input can reward retrieval of the easiest view rather than synthesis. CVBench curates verifiable single-view-insufficient items [22]; PinPoint adds multiple positives, explicit hard negatives, multi-image queries, and paraphrase testing [23]. Moreover, ViewDiag finds that predictions can remain consistent across views while remaining wrong [24]. Our context and crop are not synchronized cameras, so we do not claim geometric reasoning. Paired-view value must instead survive best-single-view comparison, assigned-versus-random deletion, paraphrases, and evidence-budget tests.

III. THE UTS-UAV BENCHMARK

A. Operational Event Graph

Given a context image x^g and a local evidence crop x^l , the pair task predicts an event set $Y \subseteq \mathcal{Y}$. Each event $y \in Y$ is represented as

$$y = (d, e, s, r, c), \quad (1)$$

where d is the operational domain, e the entity or region, s its state, r a spatial or functional relation, and c the scene context. Missing factors are marked null rather than guessed. A label definition contains positive visual criteria, exclusion rules, required evidence factors, and an escalation caveat separating visual evidence from a legal conclusion.

The resulting Operational Event Graph has two edge types. *Hierarchy edges* connect a concept to broader domain and entity nodes. *Confusion edges* connect labels that share

TABLE I

CONTENT-AUDITED DEVELOPMENT SNAPSHOT. COUNTS ARE PAIR ANNOTATIONS UNLESS MARKED AS EXACT-HASH CONTENT GROUPS.

Quantity	Value
Catalog concepts / raw records	67 / 26,749
Present / absent concepts	45 / 22
Pair annotations	9,093
Unique context / crop / pair content	7,279 / 9,056 / 9,082
Multi-evidence groups / annotations	1,389 / 3,197
Multi-label groups / annotations	269 / 623
Redundant exact pairs	11 (0.12%)
dHash review groups / annotations	223 / 644
Outside named folder date window	7,778 (85.5%)
Class-level source IDs / image joins	1,549 / 0
Verified source media / linked contexts	0 / 0
Provenance structural / scientific gate	PASS / BLOCKED
Proxy components / largest share	42 / 37.1%
Proxy 30/10/10 support concepts	4

most factors but differ in a decision-critical factor, such as floating versus waterside garbage (r), building material versus construction waste (s), or smoke emission versus suspected fire (c and temporal evidence). These neighbors define hard-negative sampling and diagnostic evaluation. Orthogonally, each unique context induces a Context–Evidence Event Graph

$$G_i = (x_i^g, \{x_{ij}^l\}_{j=1}^{m_i}, Y_i, A_i), \quad (2)$$

where Y_i is the union of valid context events and $A_i \subseteq \{1, \dots, m_i\} \times Y_i$ links evidence crops to events. This many-to-many graph defines a context-set task in addition to pair recognition.

B. Reconciled Snapshot and Release Tiers

The business catalog contains 67 concepts and 26,749 accumulated annotation records; those records are not verified unique images. A byte-level filesystem audit finds 9,093 filename-matched context/crop annotations across 45 present concepts, with no orphan or undecodable pair. Exact hashes reduce these rows to 7,279 unique contexts, 9,056 unique crops, and 9,082 unique pairs. The visible distribution remains extreme: the largest concept contributes 73.9% of pair annotations, the largest two contribute 81.3%, 41 catalog concepts have fewer than ten visible pairs, and 22 have none. Forty-five class-level download logs expose 1,549 unique business record IDs, but none occurs in the filename or pair manifest, yielding zero deterministic record-to-context joins. Table I reports the versioned audit snapshot.

We therefore avoid an indefensible 67-way supervised split and define three tiers:

- **Core.** A concept enters supervised evaluation only if it has at least 60 deduplicated source groups, at least three true acquisition groups, and enough groups to place ten or more in each scored split. No concept is finalized until flight/site provenance is recovered. A proxy 30/10/10 context-count diagnostic is reported only as split feasibility, never as Core eligibility.

- **Tail.** Present concepts that fail the core gate are retained for text-only zero-shot, definition-plus-exemplar few-shot, retrieval, and data-acquisition evaluation. They are never oversampled into a conventional test set.
- **Catalog.** Concepts without reconciled or releasable images remain ontology nodes and collection targets, not scored classes. This prevents taxonomy ambition from being misreported as benchmark coverage.

C. Context–Evidence Sets and Evidence Cards

For each pair annotation, x^g preserves the wide inspection context and x^l is a linked crop containing putative local evidence. Exact-content grouping reveals 1,389 contexts with multiple evidence crops (3,197 pair annotations) and 269 contexts with multiple labels (623 annotations). Only 11 redundant context–crop pairs exist; thus most repeated contexts encode useful multi-instance or multi-event structure rather than file duplication. One byte-identical pair appears under two labels and is held out pending adjudication; if both labels are valid, it becomes a multi-label target rather than contradictory supervision.

The pair track enables context-only, crop-only, and joint inputs. The set track receives one context and all linked oracle crops, then predicts Y_i and the evidence assignment A_i . At deployment, crops may come from an existing detector or retrieval proposal; both tracks separately report oracle- and proposed-crop performance so localization failure is not hidden inside recognition.

We instantiate the remaining uncertainty as four deterministic, row-preserving review ledgers: 623 annotations in 269 multi-label context cases; 644 members in 223 nonexact near-duplicate cases; two annotations in the one exact context–crop cross-label conflict; and one provenance-recovery record for each of the 7,279 exact contexts. Hashes select cases but never decide semantics or provenance. Controlled vocabularies separately encode annotation actions, evidence support, duplicate relations, and provenance status. Two isolated reviewer assignments expose the linked images in an offline visual interface; a strict merger checks identity, row coverage, immutable fields, and controlled values before generating an adjudicator-signed ledger. Agreement is never promoted to ground truth automatically, and excluded rows remain so every frozen manifest can be reproduced.

Every released sample is accompanied by an evidence card with: canonical event and five factors; source, flight, location, and time group identifiers when recoverable; crop-to-source linkage; positive evidence; excluded alternatives; required evidence type (object, region, state, relation, context, or temporal); and review status. A small, stratified grounding subset receives a human-checked box or polygon. Samples whose policy meaning cannot be inferred from a still image are marked `needs_temporal_evidence` and excluded from still-image claims.

D. Caption Construction and Audit

For a verified pair (x^g, x^l, y) , a teacher VLM first produces a label-agnostic description $c^{(0)}$ of visible content. A

TABLE II

POSITIONING AGAINST REPRESENTATIVE AERIAL RESOURCES. SCALE ALONE IS NOT OUR CLAIM; UTS-UAV CONTRIBUTES PAIRED CONTEXT/EVIDENCE, OPERATIONAL FACTORIZATION, AND LEAKAGE-AWARE GROUPS. COUNTS FOLLOW THE CITED PAPERS AND OUR RECONCILED DEVELOPMENT SNAPSHOT.

Dataset	Scale	Real	Primary scope	Paired context/evidence	Operational factors
VisDrone [5]	10,209 images	Yes	detection and tracking	No	No
ERA / CapERA [8], [9]	2,864 videos	Mixed	25 events and captions	No	No
UAV-Human [6]	67,428 videos	Yes	human behavior	No	No
UAVBench [1]	261k images	Yes	10 image/region VLM tasks	Region annotations	No
SpatialSky [2]	1M samples	–	13 UAV spatial categories	Multi-granularity	No
UAVLight [19]	repeat paths	Yes	illumination-robust 3D reconstruction	Multi-time views	No
UAVReason [3]	273k samples	No	22 spatial/temporal reasoning types	Multi-frame	No
UTS-UAV (ours)	9,093 pairs / 7,279 contexts	Yes	operational event sets	Yes	ontology + sample graph

constrained rewrite then receives $c^{(0)}$, the event definition, exclusions, and verified factors to produce a concise evidence statement c . The rewrite must name the event, cite only visible evidence, distinguish the nearest confusion label, and state uncertainty when a required factor is not visible.

Automatic gates reject missing event phrases, factor contradictions, unsupported named entities, and low image–text entailment. A counterfactual audit replaces y with graph neighbor y^- ; a captioner that invents support for y^- is flagged. Human review covers every core test item, every rare/tail item used for evaluation, and a stratified sample of training captions. We will report acceptance, correction, and counterfactual-susceptibility rates rather than treating synthetic text as ground truth by default.

E. Leakage-Resistant Splits and Versioning

Exact hashes, perceptual hashes, and embedding-neighbor search identify duplicates and near duplicates. The exact-context group is the minimum indivisible split unit; all 223 nonexact dHash candidate groups (644 rows) require review and conservatively remain in one partition. Content identity and acquisition identity are separate axes: exact deduplication cannot rule out flight- or site-level shortcuts [21]. Final groups are formed in decreasing priority from original media identifier, flight, location, and time window, with every crop and its context together. Entire groups are assigned to train, validation, seen-location test, or unseen-location test. Hyperparameters are frozen on validation data, and the test set is evaluated once per seed.

We make the unresolved gap measurable through a four-level provenance ladder: A requires verified source media, flight, site, time, and permission; B retains verified media and time but incomplete geography; C contains filename-date and content-group proxies only; and D has no parseable date. All current contexts are C or D. For pipeline debugging, a provisional split unions contexts sharing an inferred date or dHash candidate group, producing 42 indivisible components with zero exact, near, or inferred-date cross-split hits. This is still unusable for main results: the largest component holds 2,691 contexts (37.1%), only four concepts meet a weak 30/10/10 context-support diagnostic, and 47/44 catalog

concepts have zero validation/test contexts. These numbers are evidence that filename dates are too coarse, not evidence of leakage-free evaluation.

Revision 6 converts that warning into a fail-closed release gate with three normalized grains: 1,549 business source records, a canonical source-media acquisition registry, and 7,279 exact context-to-media dispositions. The frozen templates pass structural validation, but the scientific gate is BLOCKED: 0/1,549 records are resolved, the media registry has 0 verified rows, and 0/7,279 contexts are linked or excluded. The official split builder therefore emits no manifest. It becomes executable only when every record and context is dispositioned, linked contexts resolve to reviewed media with flight/site/time/permission/privacy evidence, and at least four sites and eight flights exist. A synthetic complete fixture exercises the READY path and asserts zero source-media, flight, site, exact-content, and reviewed-near-group leakage; it is a software test, never dataset evidence.

The release includes a machine-readable data card, taxonomy version, split manifest, file checksums, duplicate-cluster identifiers, adjudication ledgers, and evaluation code. Any later data addition creates a new version; test groups are never silently moved into training. Sensitive coordinates and identifiers are removed or coarsened, and image release is conditional on permission and privacy review.

IV. THE CLEAR BASELINE

A. Dual-View Structured Prediction

CLEAR instantiates a deliberately reproducible baseline rather than a new foundation model. A Qwen3-VL-8B-Instruct student [18] receives the global context x^g and local evidence x^l as two ordered images with explicit CONTEXT and EVIDENCE markers. The pair-level target is

$$s = [\text{events} : Y; \text{factors} : \{(d, e, s, r, c)\}_{y \in Y}; \text{evidence} : q; \text{uncertain} : u], \quad (3)$$

where q is a short visible-evidence statement and u is a binary abstention field. Language-model and projector LoRA modules are trained while the vision tower and base language weights remain frozen. Single-view baselines use the same prompt and token budget.

Auditable data engine

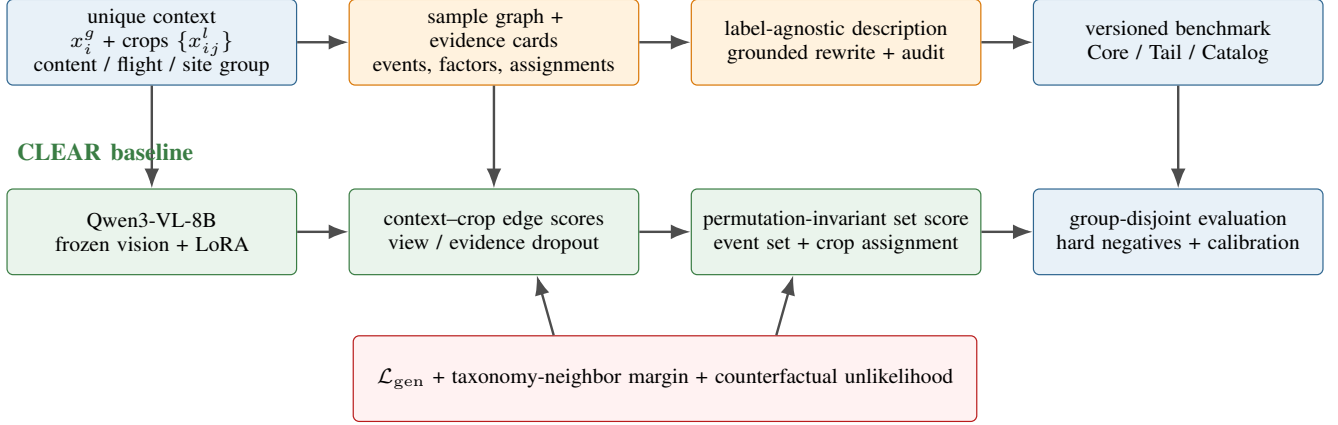


Fig. 2. UTS-UAV separates an auditable data engine (top) from CLEAR/CLEAR-Set and frozen evaluation (bottom). Exact-context grouping converts pair rows into variable-size sample graphs; edge scores are aggregated without crop order while retaining evidence assignments.

Training uses verified event phrases, factors, and audited evidence statements. The model is never given a candidate list in the free-generation setting. In the closed-set setting, the normalized conditional log-likelihood of each canonical event definition gives a reproducible label-wise score

$$S_\theta(y | x^g, x^l) = \frac{1}{|a_y|} \sum_{t \in a_y} \log p_\theta(a_{y,t} | a_{y,<t}, x^g, x^l), \quad (4)$$

where a_y is the canonical label plus its compact visual definition. Temperature-scaled label scores are thresholded independently, allowing legitimate co-occurrence rather than forcing a softmax winner.

B. Permutation-Invariant Set Extension

For a context set G_i with m_i crops, CLEAR evaluates every context-crop edge and aggregates evidence without crop order:

$$S_\theta^{\text{set}}(y | G_i) = \tau \log \sum_{j=1}^{m_i} \exp(S_\theta(y | x_i^g, x_{ij}^l) / \tau). \quad (5)$$

Binary cross-entropy over Y_i trains set prediction, while the maximizing edge supplies a transparent evidence assignment $\hat{A}_i$. We compare log-sum-exp with max pooling and a parameter-matched attention pool. This *CLEAR-Set* extension adds no crop-order encoding and is scored only on adjudicated multi-evidence groups.

C. Graph-Structured Supervision

Token-level cross-entropy alone rewards fluent background description and ignores which confusion was resolved. We retain label-span weighting from conventional grounded caption tuning, but add two graph-aware terms. For each positive $y \in Y$, a neighbor $y^- \in \mathcal{N}(y) \setminus Y$ shares the entity or scene while differing in one decision-critical factor. A margin loss requires

$$\mathcal{L}_{\text{nbr}} = \max(0, m - S_\theta(y | x^g, x^l) + S_\theta(y^- | x^g, x^l)). \quad (6)$$

We also construct a counterfactual evidence statement q^- by changing only the disputed factor. Its token probability is penalized with an unlikelihood objective

$$\mathcal{L}_{\text{cf}} = - \sum_{t \in q^-} \log(1 - p_\theta(q_t^- | q_{<t}^-, x^g, x^l)). \quad (7)$$

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{gen}} + \lambda_{\text{nbr}} \mathcal{L}_{\text{nbr}} + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}}, \quad (8)$$

where $\mathcal{L}_{\text{gen}}$ upweights canonical label tokens by γ . Ablations remove each term and replace graph neighbors with random negatives to test whether gains come from decision-boundary structure.

D. Context-Evidence Reliance

Joint input can still ignore one view. We therefore train with controlled view dropout: context-only, evidence-only, and joint examples appear with fixed probabilities. The target stays unchanged only when the remaining view satisfies the event’s required evidence factors; otherwise *uncertain* is set true. At evaluation, four conditions isolate the perception pipeline: context-only, oracle crop, context plus oracle crop, and context plus a detector/retriever proposal. The oracle-to-proposal gap measures localization error, while the joint-to-best-single-view gap measures genuine complementarity.

Accepting two images does not prove cross-view synthesis [22]. We therefore add three necessity diagnostics on adjudicated graphs. *Single-view insufficiency success* counts items that are correct jointly but incorrect from either view alone. *Assigned-evidence deletion* removes the human-linked crop for an event and compares the score drop with deletion of a size-matched random crop. Finally, a crop-budget curve reports set F1 as the model admits its top- k evidence edges. A method that cherry-picks one easy view will have a small assigned-versus-random deletion gap and a flat joint advantage even if its headline set F1 is high.

E. Confidence and Selective Prediction

Closed-set label scores are temperature-scaled on validation groups. The model abstains when no label exceeds its validation-selected threshold or when factor outputs violate the Operational Event Graph. We report classwise expected calibration error and risk–coverage curves; abstention is never counted as correct. Free-generation outputs are mapped to the taxonomy by exact canonical/synonym matching, followed by a frozen embedding tie-breaker, with all unmatched phrases counted as errors.

V. EXPERIMENTS

A. Questions and Baselines

The machine-readable experiment registry freezes four questions. **RQ1:** How far are general and aerial VLMs from operational event recognition? **RQ2:** Do paired and set-valued inputs outperform either view alone because they use complementary evidence rather than cherry-pick one view? **RQ3:** Do graph neighbors and counterfactual rejection improve decision-boundary accuracy beyond label-grounded captions? **RQ4:** Are improvements preserved across frozen acquisition groups and under selective prediction, and—only after site recovery—on unseen locations? Registry rows for main results remain blocked until the provenance gate is **READY**; only parser and training smoke tests may use the proxy split.

Zero-shot baselines include OpenCLIP ViT-L/14 [25], GeoChat [10], and Qwen3-VL-8B [18], each with direct and definition prompts. Adapted baselines use the same Qwen3-VL student, images, and split: linear probe, label-only LoRA, generic-caption LoRA, grounded-caption LoRA, two-image concatenation, and full CLEAR. If compatible, a UAVIT-1M-adapted model is evaluated without retraining on benchmark test sources. No baseline is removed after test inspection.

B. Metrics and Statistical Protocol

Pair recognition uses macro-F1 as the primary metric, with label mAP, micro-F1, per-domain macro-F1, and worst-class recall as secondary measures. Context-set prediction reports macro/micro F1, label mAP, exact-set accuracy, and evidence-assignment accuracy on adjudicated graphs. The ontology is evaluated with factor exact match, hierarchy-consistent F1, and confusion-edge pairwise accuracy. We report the view conditions from [Section IV](#), single-view-insufficiency success, assigned-versus-random deletion gaps, crop-budget area under the F1 curve, and challenge slices. Prompt robustness reports the maximum and standard deviation across six meaning-preserving paraphrases [23]; evidence sensitivity reports prediction change after assigned-evidence removal, not consistency alone [24]. Unseen-location macro-F1 is enabled only after an official site-disjoint split exists. Parsing, prompts, splits, and seeds are frozen.

Calibration uses 15-bin expected calibration error and area under the risk–coverage curve (AURC). Explanation quality is evaluated by event correctness, support by visible evidence, contradiction rate, and hallucination rate; CIDEr is omitted from the primary analysis because lexical overlap is not

evidence faithfulness. Three blinded raters score a stratified test subset, with inter-rater agreement and adjudication rate reported.

All model results are mean $\pm$ standard deviation over three seeds. Paired group bootstrap confidence intervals use 10,000 resamples at the location/flight-group level. The date/content proxy split is permitted only for parser, dataloader, and training smoke tests; its scores cannot enter the main table. If acquisition provenance remains incomplete, location generalization is removed rather than approximated with exact-context resampling. The primary comparison (CLEAR versus grounded-caption LoRA) uses a paired permutation test with Holm correction for prespecified secondary comparisons. Model selection uses validation macro-F1 and AURC; test groups are evaluated only after configuration freeze.

C. Implementation and Data Budgets

The primary student is Qwen3-VL-8B-Instruct. Context and crop images preserve aspect ratio with a shared visual-token budget; allocation is ablated rather than independently increasing compute for the joint model. The initial LoRA configuration uses rank $r = 16$, $\alpha = 32$, dropout 0.05, AdamW, and three epochs. Learning rate, token budget, view-drop probabilities, γ , margin m , and λ values are selected on validation groups from a small prespecified grid. The final paper will report exact hardware, peak memory, training energy/time, images per second, and trainable parameters.

To test whether the method survives realistic curation limits, learning curves use 1, 4, 16, and all training groups per class. The dominant class is capped per epoch; no synthetic duplication is used in validation or test. Pair batches and complete context sets are sampled separately so two crops from one context cannot masquerade as independent evidence. Tail evaluation uses text-only zero-shot and 1-/4-shot exemplars without mixing tail test images into supervised core training.

D. Ablations, Stress Tests, and Error Analysis

The ablation matrix isolates paired views, audited captions, label-token weighting, graph neighbors, counterfactual unlikelihood, and view dropout. Set ablations compare max, log-sum-exp, and parameter-matched attention aggregation, plus shuffled or missing evidence crops. Neighbor negatives are compared with frequency-matched random negatives. Crop experiments compare oracle crops, detector/retriever proposals, spatially jittered crops, irrelevant same-scene crops, human-assigned evidence deletion, and size-matched random deletion. Counterfactual tests swap only one factor, including water location, material state, coverage state, and scene relation.

Error analysis reports a confusion-edge heat map and a fixed gallery of successes, abstentions, unsupported rationales, and leakage-like near duplicates. Every preregistered row is retained. A failed baseline is marked with the concrete hardware/software reason; no **TBD** cell may remain in the submission version.

TABLE III

REGISTERED MAIN-RESULT SCHEMA, CONDITIONAL ON A READY PROVENANCE GATE AND FROZEN ACQUISITION-AWARE SPLIT. LOWER IS BETTER FOR AURC. NO PLACEHOLDER IS A RESULT.

Method	Adapt.	Views	Macro F1	mAP	Hard-neg.	Set F1	Assign.	Unseen F1	AURC ↓
OpenCLIP ViT-L/14	zero-shot	context	TBD	TBD	TBD	–	–	TBD	TBD
GeoChat-7B	zero-shot	context	TBD	TBD	TBD	–	–	TBD	TBD
Qwen3-VL-8B + definition	prompt	context	TBD	TBD	TBD	–	–	TBD	TBD
Grounded-caption Qwen3-VL	LoRA	context	TBD	TBD	TBD	TBD	–	TBD	TBD
Two-view concatenation	LoRA	pair	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CLEAR/ CLEAR-Set (ours)	LoRA	pair / set	TBD	TBD	TBD	TBD	TBD	TBD	TBD

TABLE IV

ACCURACY-EFFICIENCY ON THE SAME QWEN3-VL BACKBONE, HARDWARE, AND VISUAL-TOKEN BUDGET.

Trainable modules	Params.	Peak GB	pair/s	Macro-F1
Linear probe	TBD	TBD	TBD	TBD
Projector only	TBD	TBD	TBD	TBD
LLM LoRA	TBD	TBD	TBD	TBD
Projector + LLM LoRA	TBD	TBD	TBD	TBD
QLoRA	TBD	TBD	TBD	TBD
Full fine-tuning	TBD	TBD	TBD	TBD

VI. LIMITATIONS, ETHICS, AND RELEASE

The current asset is not yet a release-ready 67-class dataset. Raw catalog counts mix annotation records with files, 22 concepts have no visible workspace images, and pair annotations are dominated by one category. The content audit resolves pairing and exact duplication, but flight/site provenance remains absent: 1,549 business records yield zero deterministic image joins, the canonical media registry is empty, and all 7,279 exact contexts remain undispositioned. Moreover, 7,778 filename-inferred dates fall outside the folder’s advertised window and 223 nonexact dHash groups await review. The conservative proxy split has a 37.1% largest component and only four classes meeting a weak 30/10/10 support diagnostic. The executable gate therefore forbids official splits, Core classes, main scores, and unseen-location claims. The paper should be withdrawn if provenance and review gates are not met in time.

Operational labels are geographically and policy specific. A still image can provide evidence for human review but cannot by itself prove a legal violation. Categories that require persistence, intent, distance, or motion are marked as requiring temporal evidence and must not be evaluated as certain still-image events. Predictions support authorized inspection; they do not replace it.

Teacher-generated captions may rationalize an incorrect label. Label-agnostic description, factor checks, counterfactual auditing, and human review reduce but do not eliminate this risk. The 269 exact contexts carrying multiple labels, including one byte-identical context–crop pair stored under two labels, require explicit adjudication; legitimate co-occurrence

must remain multi-label rather than becoming contradictory targets. We will release synthetic-text provenance and correction statistics. Model-generated evidence is never used to repair a wrong prediction during evaluation.

UAV imagery can expose people, vehicles, homes, and locations. Public release requires authorization, removal or coarsening of coordinates and identifiers, and blurring or exclusion where appropriate. The license, geographic scope, prohibited uses, and retention policy remain **TBD**. Images shown in the draft are internal examples and must be cleared before submission.

Finally, paired crops may reflect an upstream detector and therefore encode selection bias. We expose this through context-only, oracle-crop, proposed-crop, jitter, and irrelevant-crop tests, but a benchmark built from selected inspection events cannot estimate natural event prevalence. Negative flights and non-event scenes must be added before deployment claims are made.

VII. CONCLUSION

We reframed UTS-UAV from a flat 67-class inventory into a context–evidence benchmark for compositional operational events. Its Operational Event Graph represents label meaning, while the recovered Context–Evidence Event Graph preserves multiple crops and events per scene instead of pretending that 9,093 annotations are independent frames. CLEAR provides a graph-aware VLM baseline and set extension with calibrated abstention. Revision 6 makes the acceptance condition executable: the official split cannot exist until normalized provenance is complete. Adjudication, a READY gate, the release audit, and all registered three-seed results must be completed without changing the frozen protocol.

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TABLE V

PRESPECIFIED CLEAR ABLATIONS. RANDOM NEGATIVES ARE FREQUENCY-MATCHED; ALL OTHER SETTINGS ARE FIXED.

Variant	Context	Crop	Graph neighbor	Counterfactual	Macro-F1	Pair acc.	AURC ↓
Label-only LoRA	Yes	No	No	No	TBD	TBD	TBD
Grounded-caption LoRA	Yes	No	No	No	TBD	TBD	TBD
Two-view concatenation	Yes	Yes	No	No	TBD	TBD	TBD
Random-negative margin	Yes	Yes	Random	No	TBD	TBD	TBD
Graph-neighbor margin	Yes	Yes	Yes	No	TBD	TBD	TBD
CLEAR without view dropout	Yes	Yes	Yes	Yes	TBD	TBD	TBD
CLEAR (full)	Yes	Yes	Yes	Yes	TBD	TBD	TBD

TABLE VI

STRESS-TEST MACRO-F1. “PROPOSAL” REPLACES THE ORACLE CROP WITH AN UPSTREAM DETECTOR/RETRIEVER OUTPUT.

Method	All	Small evidence	Adverse capture	Complex bg.	Unseen site	Proposal crop
Qwen3-VL + definition	TBD	TBD	TBD	TBD	TBD	TBD
Grounded-caption LoRA	TBD	TBD	TBD	TBD	TBD	TBD
Two-view concatenation	TBD	TBD	TBD	TBD	TBD	TBD
CLEAR	TBD	TBD	TBD	TBD	TBD	TBD

TABLE VII

BLINDED EVIDENCE-STATEMENT EVALUATION ON A STRATIFIED SUBSET.

Method	Event correct	Evidence supported	Contrad. ↓	Halluc. ↓
Zero-shot Qwen3-VL	TBD	TBD	TBD	TBD
Generic-caption LoRA	TBD	TBD	TBD	TBD
Grounded-caption LoRA	TBD	TBD	TBD	TBD
CLEAR	TBD	TBD	TBD	TBD

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